INCH-POUND
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SUPERSEDING
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PERFORMANCE SPECIFICATION

FELT, HAIR OR WOOL: MILDEW RESISTANT AND MOISTURE RESISTANT TREATMENT FOR

This specification is approved for use by all Departments and Agencies of the Department of the Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements for mildew resistant and moisture resistant treatment for hair and wool felt.

1.2. Classification. The finished felt treatment will be of the following types as specified.

Type I - Mildew resistant
Type II - Moisture resistant
Type III - Mildew and moisture resistant

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed: ATTN: DLA Troop Support, 700 Robbins Street, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil

AMSC N/A FSC 8305

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2.2. Government documents.

2.2.1 Specifications, standards, and handbooks. N/A

2.2.2 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Federal Insecticide, Fungicide and Rodenticide Act (FIFRA)
(40 CFR Part 162) State Registration of Pesticide Products

Federal Insecticide, Fungicide and Rodenticide Act (FIFRA)
FIFRA as amended by the Food Quality Protection Act of 1996 and the
Pesticide Registration Improvement Act of 2003

(Copies of these documents are available online at https://www.epa.gov/pesticides.)

FEDERAL TRADE COMMISSION

Rules and Regulations under the Textile Fiber Products Identification Act

(Copies are available online at https://www.ftc.gov.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILES CHEMISTS AND COLORISTS (AATCC)

AATCC Test Method 30 - Antifungal Activity, Assessment on Textile Materials: Mildew and Rot Resistance of Textile Materials
AATCC Test Method 70 - Water Repellency: Tumble Jar Dynamic Absorption Test

(Copies are available online at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY

ASQ/ASNI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies are available online at https://asq.org.)

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ASTM INTERNATIONAL

ASTM D1776 - Standard Practice for Conditioning and Testing Textiles
ASTM D1777 - Standard Test Method for Thickness of Textile Materials
ASTM D3776/D3776M - Standard Test Method for Mass per Unit Area (Weight) of Fabric
ASTM D5035 - Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Strip Method)
ASTM D2475 - Standard Specification for Felt
ASTM D5430 - Standard Test Method for Visually Inspecting and Grading Fabrics

(Copies are available online at https://www.astm.org.)

INFORMA HEALTHCARE

Repeat Insult Patch Test – Modified Draize Procedure
Principles and Methods of Toxicology, A. Wallace Hayes (editor)

(Copies are available online at https://www.crcpress.com)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. First article. When specified (see 6.2) a sample of the treated felt shall be subject to first article inspection (see 4.2).

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Material. The felt to be treated in accordance with this document shall conform to the requirements of the felt specification cited in the applicable acquisition document (see 6.1).

3.4 Color. The color of the treated, undyed felt shall be that resulting from the color imparted by the addition of the mildew or moisture resistant finishing treatment. When dyed felt is specified, the color of the treated felt shall be that resulting from the combination of the base color and the color imparted by the applied finish.

3.5 Finishing treatment.

3.5.1 Mildew resistant treatment (Type I). The felt shall be treated with an organic mildew inhibitor that is EPA registered under the FIFRA and it shall retain a 50 percent breaking strength after 12 weeks of soil burial exposure after leaching. The observable fungi shall be no greater

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than “Microscopic Growth” (visible only under the microscope) after leaching when tested as specified in 4.5.

3.5.2 Moisture resistant treatment (Type II). The felt shall be treated with a moisture resistant compound and it shall not absorb more than 50 percent in weight, based on the conditioned weight of the sample when tested as specified in 4.5.

3.5.3 Mildew and moisture resistant treatment (Type III). The combined mildew and moisture resistant treatment shall conform to the requirements as specified for Type I and Type II (see 3.5.1 and 3.5.2) when tested as specified in 4.5

3.5.4 Finished weight and thickness. The increase in weight and thickness of the finished felt shall not exceed 15 percent for Type I and II and 20 percent for Type III based on the maximum specified weight or thickness for the untreated felt when tested as specified in 4.5.

3.6 Width. For Government procurements only, the width of the felt shall be as specified in the applicable felt specification (see 6.1).

3.7 Toxicity. The finished felt shall not present a health hazard and shall show compatibility with prolonged direct skin contact when tested as specified in 4.5.3. The use of chemicals recognized by the EPA as human carcinogens is prohibited.

3.8 Length and put-up. Unless otherwise specified (see 6.2), (for Government procurements only), the felt length shall as specified in the felt specification.

3.9 Fiber identification. Each roll or sheet of the finished felt shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.10 Marking. Each roll or sheet shall be identified as specified in the felt specification.

3.11 Workmanship. The finished felt shall conform to the quality of product established in this specification. The occurrence of defects as specified in 4.4.shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.1.1 Inspection conditions. Unless otherwise specified, excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

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4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5.

4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.4 Visual Examination. The visual examination and the defects assigned to the treated felt shall be as specified in the applicable specification for the untreated felt except the defects specified in Table I shall be scored four (4) points for each treated yard or sheet in which they occur when evaluated in accordance with ASTM D5430.

TABLE I. Defects that score four (4) points for treated felt.

Defect Description
Clearly visible crystallization of treatment on surfaces of felt
Non-uniform application of treatment
Non-uniform color resulting from uneven treatment
Objectionable odor.
Any embedded foreign matter or uncleanliness.

4.5 Methods of testing and inspection.

4.5.1 End item testing. The felt shall be tested for the characteristics listed in Table II. The methods of testing specified, wherever applicable and as listed in Table II shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) sheets or five (5) continuous yards full width of the finished felt for all physical tests. The lot shall be unacceptable if one (1) or more sample units fail to meet any requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE II. Finished felt tests.

Characteristic	Requirement Reference	Test method
Rot resistance, Type I and III, (percent strength retained), (min.) 1/ After leaching, 6/	3.5.1 and 3.5.3	AATCC 30, Test I, 3/, 4/
Mildew resistance, Type I and III, (observed fungus growth),(max.) 2/ After leaching 6/	3.5.1 and 3.5.3	AATCC 30, Test III, 5/ (see 6.4)
Moisture resistance, Type II and Type III, (percent absorption) (max.)	3.5.2 and 3.5.3	4.5.2
Weight, oz./sq. yd.	3.5.4	ASTM D D3776/D3776M
Thickness	3.5.4	ASTM D1777
Toxicity	3.7	4.5.3

1/ AATCC 30, Test I will be required for First Article testing only with technical approval from the Procuring Activity.
2/ AATCC 30, Test III will be used for First Article and subsequent lot testing with technical approval from the Procuring Activity
3/ The samples shall be evaluated for breaking strength using ASTM D5035 before soil burial exposure and after leaching and soil burial exposure to calculate the percent strength retained from soil burial exposure. It shall be calculated as follows:

Percent strength retained = $100-((O-F/O)\ x100)$

Where:
O = Original (treated) breaking strength before soil burial exposure
F = Final breaking strength after leaching and solid burial exposure

4/ Five (5) specimens shall be evaluated for each the machine and cross-machine direction.
5/ Five (5) specimens shall be evaluated.
6/ Leaching shall be conducted in accordance with procedure in Appendix A1 of AATCC 30, prior to testing.

4.5.2 Moisture resistant test. Three (3) test specimens 3-inch by 3-inch shall be tested. The specimens shall be conditioned in accordance with ASTM D1776. The conditioned specimens shall be rolled together and weighed to the nearest 0.01 grams. This is the “Original” weight and shall be designated as “O”. The specimens shall be completely immersed in a vessel with distilled water for 20 minutes at a temperature of 80 (±) 3°F. At the end of the immersion period, specimens shall be removed from the water and each specimen shall be placed between two sheets of blotting paper and passed once though a wringer. Blotting paper and wringer shall conform to AATCC Test Method 70 requirements. The specimens shall be removed quickly from the blotting paper, rolled together and weighed to the nearest 0.01 grams. This is the “Final” weight and shall be designated as “F”.

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The percent moisture absorbed shall be calculated as follows:

Percent = $(F - O/O) \times 100$

Where O = Original
F = Final

4.5.3 Toxicity test. When required, (see 6.2) an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the material is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If toxicity requirement (see 3.7) can be demonstrated with historical use data, toxicity testing may not be required (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The treatment is intended to be applied to felt covered by MIL-DTL-32065, Felt Sheet (Hair) and Felt Roll (Hair) and ASTM D2475 Standard Specification for Felt (see Note below). The treatments do not apply to felts used on medical items.
NOTE: ASTM D2475 Standard Specification for Felt replaced C-F-206, Felt Sheet: Cloth, Felt, Wool, Pressed. The ASTM standard does not reference when a mildew treatment is required. May need to check C-F-206 to determine which classes require mildew treatments.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of felt and treatment required (see 1.2 and 3.5).
c. The specific issue of individual documents referenced (see 2.2).
d. When first article is required (see 3.1)
e. The applicable felt specification (see 3.3)
f. Conformance inspection acceptance quality limits (AQL) (see 4.3)
g. Inspection conditions (see 4.1.1)
h. When toxicity testing is required (see 3.7 and 4.5.3)
i. Packaging (see 5.1)

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6.3 First article. When first article inspection is required (see 3.1), it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.5 Subject term (key word) listing.

Immersion test
Microscopic growth
Rot resistance
Soil burial

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - GL
Navy – NU
Air Force – 11

Preparing activity:
DLA-CT

Review Activities:
Navy – MC

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